1] What is the role of optimizer?

It's primary role is to minimize the model's error or loss function, enhancing Performance.

i] Calculate updates for model Parameters

ii] minimize the loss function

(Reduce the Prediction error improve model accuracy)

iii] Control learning speed

iv] Improves Convergence.

(optimal solution faster more accuracy)

v] Avoid the Problem like local minima and oscillation.

a] Types of gradient Descent approach:

1] Batch gradient Descent approach:

gradients using the entire dataset in each iteration.

Advantage:

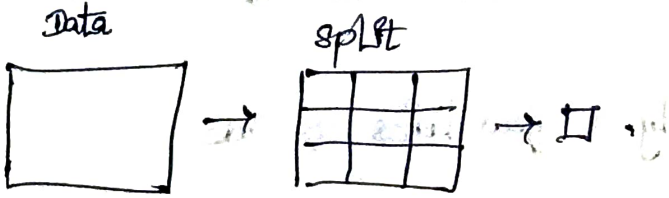
→ Steady and accurate Convergence to a minimum
→ Easy Implementation
→ Excellent for Convex function (→ graph of function, minimum point)

Disadvantage:

→ Impractical for large dataset.
→ Slow to Converge
→ Difficult to Escape Local minima.

2] Mini-batch gradient Descent:

Combines Batch and SGD by using small batches of data for updates



- large to split in minimum.

Advantage:

→ Provide greater accuracy than stochastic gradient Descent.
→ Able to handle large Dataset.
→ " " escape Local minima.

Disadvantage:

→ more Complex Algorithm.
→ Additional Hyper-parameter
→ less accurate than batch gradient Descent.

3] Stochastic gradient Descent: (use for Rms Prop)

Uses on data point Per iteration to Compute gradients making it faster.

Advantage:

→ Faster Convergence - suitable for large Dataset
→ Able to escape local minima
→ no Additional Hyperparameter

Disadvantage:

→ less accurate Convergence
→ more Complex implementation
→ non - deterministic.

3] Types of optimizer. (Advantage & disadvantage).

√1] Gradient Descent:

AD:

→ GD is a very efficient algorithm for optimizing model with a large number of Parameter.

→ Algorithm can be applied to different type of models.

→ Requires only a few lines of code.

DA:

→ local minima.

→ Requires tuning (Finding optimal hyperparameter Can be time Consuming and require a lot of trail and error.

→ Sensitive feature scaling.

2] Stochastic:

Ad:

→ Faster Convergence
→ Lower memory requirements
→ Ability to handle noisy data.
→ Generalization

Disadvantage:

→ Possibility of getting stuck in local minima.
→ Need for Careful tunning.
→ Sensitivity to initialization.
→ may require more iteration.

✓3. Adam:

Ad:

→ Combines the benefits of momentum and Adagrad.
→ Adapts the learning rate for each parameter based on both the first and second moments of their Past gradients.
→ works well for a wide range of Problems and architectures

DA:

→ may suffer from overfitting when used with smaller dataset.

4. Adagrad:

Ad:

→ Each Parameter based on the sum of the Squares of its Past gradient, suitable for sparse dataset.
→ Perform well for low learning Rate.

DA:

→ learning rate become too smaller over time.
→ Require more memory than SGD to the need to store Past gradients sum.

5. Rms Prop:

Ad: → steps to control
→ Perform well various domain → Handle Horizontal.
→ Non-stationary to control. → Handle different learning rate.

D. A:

→ may Converge Slowly Compared to other optimizer

→ not suitable for Problems with sparses data

6. Adadelta:

AD:

→ Address the small learning rate problem by using an exponentially decaying average of Past gradient.

→ require less memory. only need to Store Past gradient.

D. A:

→ learning rate Can become too Smaller to overtime

→ many Converge more slowly than other optimizer.

1] Regularization: Model romba training extra add agama pathuta,

Type:

L1 - Lasso - All weights to 0.

L2 - Ridge, - large weight to reduce.

Ad:

→ Reduces overfit

→ Prevent model from becoming too Complex.

→ Improve generalization of unseen data.

DA:

→ Choosing the regularization parameter (λ) is difficult

→ may increase traning time slightly.

→ Too much regularization Cause overfitting

2] Dropout:

Training time random neurons to be disabled.

Ad:

→ Improve model generalization

→ Prevents neurons from depending too much on each other

→ Simple & easy to implement.

DA:

→ Training become slower.

→ Choosing dropout rate is tricky

→ Too much dropout can reduce model accuracy.

3] Early Stopping

Ad:

→ Prevent over-training

→ Reduce overfitting

→ Saves training time and Computational Cost

D.A:

→ Requires Validation dataset

→ Choosing Patience / stop Criteria is difficult

→ may stop too early before optimal learning.